MILITARY SPECIFICATION SHEET

INSULATION SLEEVING, ELECTRICAL, HEAT SHRINKABLE, POLYOLEFIN, FLEXIBLE, CROSSLINKED

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the sleeving described herein shall consist of this document and the latest issue of MIL-I-23053.

REQUIREMENTS:

Continuous operating temperature range: Classes 1 and 2: -55°C (-67°F) to +135°C (+275°F); Class 3: -25°C (-13°F) to +125°C (+257°F).

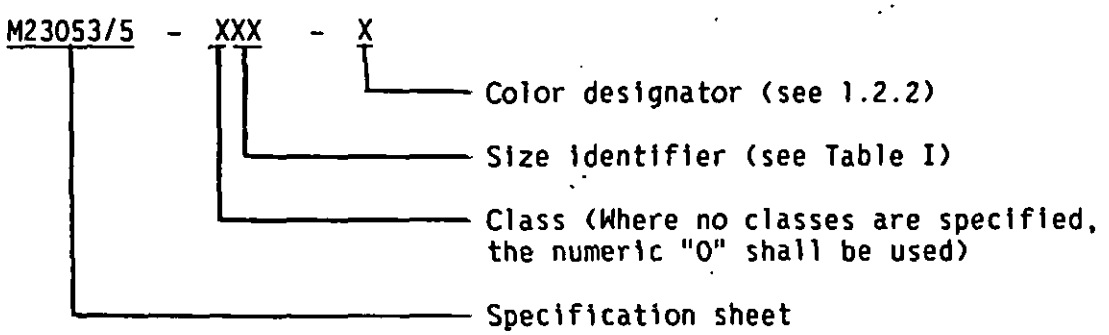
Classification: The heat shrinkable sleeving shall be furnished in the following classes, as specified (see 6.2.1):

Class 1 - Flame resistant
Class 2 - Flammable (clear only)
Class 3 - Highly flame resistant

Color: Class 2 shall be furnished in clear. Classes 1 and 3 shall be furnished in colors only. Colors shall conform to the requirements of Class 1 of MIL-STD-104.

Longitudinal change: Classes 1 and 2: ± 5 percent, overexpanded sleeving +5, -50 percent; Class 3: +1, -10 percent.

Military part number: The Military part number shall consist of the basic number of this specification sheet and dash numbers as follows:



NOTE: Entire specifiction sheet revised.

AMSC N/A FSC 5970

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Example: Class 1, gray (slate), 0.250 inch (6.35 mm) as supplied ID sleeving shall be identified as M23053/5-106-8.

TABLE I. Construction details, inches (mm). 1/

Military part number 3/	As supplied ID minimum	After unrestricted shrinkage	
		ID maximum	Wall thickness 2/
Class 1			
M23053/5-101-*	.046(1.17)	.023 (.58)	.016 ± .003 (.406 ± .076)
M23053/5-102-*	.063(1.60)	.031 (.79)	.017 ± .003 (.432 ± .076)
M23053/5-103-*	.093(2.36)	.046 (1.17)	.020 ± .003 (.508 ± .076)
M23053/5-104-*	.125(3.18)	.062 (1.58)	.020 ± .003 (.508 ± .076)
M23053/5-105-*	.187(4.75)	.093 (2.36)	.020 ± .003 (.508 ± .076)
M23053/5-106-*	.250(6.35)	.125 (3.18)	.025 ± .003 (.635 ± .076)
M23053/5-107-*	.375(9.53)	.187 (4.75)	.025 ± .003 (.635 ± .076)
M23053/5-108-*	.500(12.7)	.250 (6.35)	.025 ± .003 (.635 ± .076)
M23053/5-109-*	.750(19.1)	.375 (9.53)	.030 ± .003 (.762 ± .076)
M23053/5-110-*	1.000(25.4)	.500 (12.7)	.035 ± .005 (.889 ± .127)
M23053/5-111-*	1.500(38.1)	.750 (19.1)	.040 ± .006 (1.016 ± .152)
M23053/5-112-*	2.000(50.8)	1.000 (25.4)	.045 ± .007 (1.143 ± .178)
M23053/5-113-*	3.000(76.2)	1.500 (38.1)	.050 ± .008 (1.270 ± .203)
M23053/5-114-*	4.000(101.6)	2.000 (50.8)	.055 ± .009 (1.397 ± .229)
OVEREXPANDED			
M23053/5-115-*	1.000(25.4)	.275 (6.99)	.045 ± .007 (1.143 ± .178)
M23053/5-116-*	2.000(50.8)	.550 (14.0)	.045 ± .007 (1.143 ± .178)
M23053/5-117-*	3.000(76.2)	.810 (20.6)	.045 ± .007 (1.143 ± .178)
M23053/5-118-*	4.000(101.6)	1.050 (26.7)	.045 ± .007 (1.143 ± .178)
M23053/5-119-*	1.000(25.4)	.462 (11.7)	.045 ± .007 (1.143 ± .178)
M23053/5-120-*	2.375(60.3)	.680 (17.3)	.045 ± .007 (1.143 ± .178)
M23053/5-121-*	3.000(76.2)	.840 (21.3)	.045 ± .007 (1.143 ± .178)
M23053/5-122-*	3.750(95.3)	.930 (23.62)	.045 ± .007 (1.143 ± .178)
M23053/5-123-*	4.500(114.3)	1.450 (36.83)	.045 ± .007 (1.143 ± .178)
Class 2			
M23053/5-201-C	.046(1.17)	.023 (.58)	.016 ± .003 (.406 ± .076)
M23053/5-202-C	.063(1.60)	.031 (.79)	.017 ± .003 (.432 ± .076)
M23053/5-203-C	.093(2.36)	.046 (1.17)	.020 ± .003 (.508 ± .076)
M23053/5-204-C	.125(3.18)	.062 (1.58)	.020 ± .003 (.508 ± .076)
M23053/5-205-C	.187(4.75)	.093 (2.36)	.020 ± .003 (.508 ± .076)
M23053/5-206-C	.250(6.35)	.125 (3.18)	.025 ± .003 (.635 ± .076)
M23053/5-207-C	.375(9.53)	.187 (4.75)	.025 ± .003 (.635 ± .076)
M23053/5-208-C	.500(12.7)	.250 (6.35)	.025 ± .003 (.635 ± .076)
M23053/5-209-C	.750(19.1)	.375 (9.53)	.030 ± .003 (.762 ± .076)
M23053/5-210-C	1.000(25.4)	.500 (12.7)	.035 ± .005 (.889 ± .127)
M23053/5-211-C	1.500(38.1)	.750 (19.1)	.040 ± .006 (1.016 ± .152)
M23053/5-212-C	2.000(50.8)	1.000 (25.4)	.045 ± .007 (1.143 ± .178)
M23053/5-213-C	3.000(76.2)	1.500 (38.1)	.050 ± .008 (1.270 ± .203)
M23053/5-214-C	4.000(101.6)	2.000 (50.8)	.055 ± .009 (1.397 ± .229)

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TABLE I. Construction details, inches (mm). 1/ - Continued

Military part number 3/	As supplied ID minimum	After unrestricted shrinkage	
		ID maximum	Wall thickness 2/
Class 3			
M23053/5-301-*	.046(1.17)	.023 (.58)	.016 ± .003 (.406 ± .076)
M23053/5-302-*	.063(1.60)	.031 (.79)	.017 ± .003 (.432 ± .076)
M23053/5-303-*	.093(2.36)	.046 (1.17)	.020 ± .003 (.508 ± .076)
M23053/5-304-*	.125(3.18)	.062 (1.58)	.020 ± .003 (.508 ± .076)
M23053/5-305-*	.187(4.75)	.093 (2.36)	.020 ± .003 (.508 ± .076)
M23053/5-306-*	.250(6.35)	.125 (3.18)	.025 ± .003 (.635 ± .076)
M23053/5-307-*	.375(9.53)	.187 (4.75)	.025 ± .003 (.635 ± .076)
M23053/5-308-*	.500(12.7)	.250 (6.35)	.025 ± .003 (.635 ± .076)
M23053/5-309-*	.750(19.1)	.375 (9.53)	.030 ± .003 (.762 ± .076)
M23053/5-310-*	1.000(25.4)	.500 (12.7)	.035 ± .005 (.889 ± .127)
M23053/5-311-*	1.500(38.1)	.750 (19.1)	.040 ± .006 (1.016 ± .152)

1/ Diameter limits for object to be enclosed shall be as recommended in technical data.

2/ Wall thickness values are less when shrinkage is restricted.

3/ The asterisk in the part number shall be replaced by color code designations (see 1.2.2).

Unrestricted shrinkage: Test method 4.6.5: 200° ± 2°C (392° ± 4°F) for 3 minutes.

TABLE II. Physical properties. 1/

Characteristic	Requirement	Test procedure and conditions
As supplied:		
ID, minimum	Table I	4.6.3
Heat shock	No cracks, flowing or dripping	4.6.8 Classes 1 and 2: 250° ± 3°C (482° ± 6°F); Class 3: 225° ± 3°C (437° ± 6°F)
Secant modulus, psi (MPa), maximum	25,000 (172.4)	4.6.13.1, 2 percent strain, ASTM D 882
Concentricity	70% minimum (50% minimum for over expanded sizes)	4.6.3.3

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TABLE II. Physical properties. 1/

Characteristic	Requirement	Test procedure and conditions
Color stability	Pass	4.6.16 175° ± 2°C (347° ± 4°F), 24 hours
Clarity stability Clear	Pass	4.6.17 175° ± 2°C (347° ± 4°F), 24 hours
Restricted shrinkage	No cracks	4.6.6.1.1 175° ± 2°C (347° ± 4°F)
Voltage withstand	Pass	4.6.6.2
After unrestricted shrinkage:		
ID, maximum	Table I	4.6.3
Wall thickness	Table I	4.6.3
Tensile strength, psi (MPa), minimum	1,500 (10.3)	4.6.14 ASTM D 638, 20 inches/minute
Ultimate elongation, percent, minimum	200	4.6.14 ASTM D 638, 20 inches/minute
Dielectric strength, volts/mil (Kv/mm), minimum	500 (19.7)	4.6.2 ASTM D 2671
Volume resistivity, Ohm-cm, minimum	Classes 1 & 3 -10^{14}; Class 2 - 10^{16}	4.6.2 ASTM D 876
Dielectric constant, maximum 3/	Classes 1 and 3 - 3.1 Class 2 - 2.7	4.6.2 ASTM D 150
Specific gravity, maximum	Class 1 - 1.35 Class 2 - 1.00 Class 3 - 1.50	4.6.2 ASTM D 792
Water absorption, percent, maximum	Classes 1 and 3 - 0.5 Class 2 - 0.2	4.6.2 ASTM D 570, 24 hrs.at 23°C
Corrosion	No corrosion	4.6.10.1 and 4.6.10.2 175° ± 2°C (347° ± 4°F), 16 hours

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TABLE II. Physical properties. 1/ - Continued

Characteristic	Requirement	Test procedure and conditions
Low temperature flexibility	No cracking	4.6.7.1 Classes 1 & 2: -55° ± 1°C (-67° + 2°F); Class 3: -25° ± 1°C (-13° ± 2°F)
Flammability	Class 1 - self-extinguishing within one minute and no more than 25 percent of indicator flag burned or charred. No dripping, flowing of large sizes;	4.6.15.1
	Class 2 - N/A;	
	Class 3 - Pass	4.6.15.3
Heat resistance, property after:		4.6.9 for 168 hours, Classes 1 and 2: 175° ± 2°C (347° ± 4°F); Class 3: 150° ± 2°C (302° ± 4°F)
Ultimate elongation, percent, minimum	100	
Fluid resistance, property after:		4.6.11 (Use 23°C, not 35°C).
Tensile strength, psi (MPa), minimum	1,000 (6.9)	
Dielectric strength, volt/mil (Kv/mm), minimum	400 (15.8)	
Fungus resistance 2/	No growth	4.6.2 ASTM G 21

1/ Unless otherwise specified, the stated requirements, test conditions and procedures are for all classes.

2/ MIL-I-23053/5 materials do not normally support fungus growth. Performance of this requirement is at the option of the acquisition activity (see 6.2.1).

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3/ Dielectric constant is a requirement only when specified in acquisition document.

Shelf life conditions: Supplier shall certify to storage at 40°C (104°F) for 4 years. Conformance to 3.5. See 3.5.2 for shelf life extension.

Intended use: Heat shrinkable flexible polyolefin sleeving is used for light duty harness jackets, wire color coding, marking or identification.

NOTE:

a. Flammability - Class 2 heat shrinkable sleevings described in this specification sheet have not been flame retarded and will burn readily. These sleevings (Class 2) shall not be considered for acquisition when flame resistance is required.

Custodians
Army - ER
Navy - AS
Air Force - 85

Review activities:
Army - AR, ER, MI
Navy - EC
Air Force - 99
DLA - GS

User activities:
Army - ME
Navy - MC

Preparing activity:
Navy - AS

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